# Evangelos Gkiolas

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## EDUCATION

**University of Patras, Patras, Greece** 10/2020 – Expected 2027

- Diploma in Electrical and Computer Engineering (5-year Integrated Master; 300 ECTS)
  - Specialization: Systems, Control and Robotics
  - Relevant Coursework: Non-Linear Systems & Control, State-Space Linear Systems Analysis, Adaptive Control & Reinforcement Learning, Intelligent Control, Robotic Systems I, Artificial Intelligence I, Linear & Combinatorial Optimization, Applied Optimization
- Diploma Thesis (in progress): *"Puzzle Assembly via a Robotic Manipulator"*
  - Built an end-to-end vision and solving pipeline in Python that solves real 9- and 20-piece puzzles from piece shape alone
  - Segmentation: IS-Net (PyTorch / ONNX Runtime) on full-resolution crops with sub-pixel contour refinement; selected after comparing $U^2$-Net, IS-Net and BiRefNet (under 0.5 s per piece)
  - Shape analysis following Webster et al. (1991): skeleton-based isthmus detection of tabs and blanks, corners from the convex hull of the tab-free shape, tab/blank/flat side labelling
  - Backtracking solver (border first, then interior) that joins two sides only if they pass both an isthmus-compatibility check and a geometric check (equal length, curves agree after rigid ICP)
  - Next: backlit capture and camera calibration, then assembly with a suction-gripper robotic arm

**Politecnico di Milano (PoliMi), Milan, Italy** 02/2026 – 07/2026

- Erasmus+ exchange student (MSc level), Electrical Engineering
- Coursework: Advanced and Multivariable Control, Power Generation and Renewables, Smart Grids: Components, Functionalities and Benefits

## PROJECTS

**Adaptive Control of Nonlinear Systems – Course Project (team of 2)** 2025

- Designed and simulated adaptive controllers (2nd-order MRAC with integral action, adaptive backstepping, MIT rule) on three nonlinear benchmarks: block on an inclined plane, inverted pendulum on a cart, and ball balancing on a tilting/lifting platform
- Derived the nonlinear dynamics and linearized state-space models; Lyapunov-based adaptation laws under unknown mass parameters
- Compared controllers on tracking error and control effort (MRAC: smooth response; backstepping: faster but oscillatory input) and analyzed failure cases

**Optimal Football Lineup Selection – Binary Integer Programming (Python, Gurobi)** 2024

- Modeled starting-XI selection for Olympiacos FC (2023–24) as a BIP with 42 binary player–position variables and formation constraints; compared 5 formations
- Custom sensitivity analysis for integer programs: rating tolerance intervals per starter and optimal-replacement analysis for absent players
- Optimal 4-3-3 matched 9 of the 11 starters in the 2024 UEFA Conference League final
- Code & report: github.com/Vagkiolas/football-squad-optimization-bip

## SKILLS & LANGUAGES

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|---|---|
| **Programming:** | Python, C, MATLAB, Java |
| **Vision & ML:** | OpenCV, scikit-image, NumPy, SciPy, PyTorch, ONNX Runtime |
| **Tools:** | Simulink, ROS, Gurobi, Git, LaTeX, LTspice, AutoCAD |
| **Languages:** | Greek (native), English (C2, fluent), Italian (basic) |